

Reliability and convergence of approach/avoidance bias assessment tasks in the food consumption domain



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Abstract

Theories of motivation posit that people will more readily approach positive or appetitive stimuli, and there has been growing interest in the relationship between biases in approach and avoidance behaviours for food cues and food craving and consumption behaviour. Two paradigms commonly employed by research to investigate this relationship are the approach-avoidance task (AAT) and the stimulus-response compatibility task (SRCT). However, it is yet to be determined whether the measures yielded by these tasks reflect the same processes operating in the food craving and consumption domain. The present study examined the internal reliability and convergence of AAT and SRCT paradigms in their assessment of biased approach to unhealthy compared with healthy food stimuli, and whether the measures yielded by the AAT and SRCT paradigms demonstrated associations with individual differences in food craving and eating behaviour. One hundred twenty-one participants completed an SRCT, an AAT using an arm movement response mode, and an AAT using a key-press response mode. The measures yielded comparable and acceptable levels of internal consistency, but convergence between the different task bias scores was modest or absent, and only approach bias as measured with the AAT task using an arm movement response mode was associated with self-report measures of eating behaviour and trait food craving. Thus, tasks did not converge strongly enough to be considered equivalent measures of approach/avoidance biases, and the AAT task using an arm movement response seems uniquely suited to detect approach biases argued to characterise maladaptive eating behaviour and craving.

Keywords

Approach/avoidance; eating; approach-avoidance task; stimulus-response compatibility task; food consumption; action tendencies

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Theories of motivation posit that people will more readily approach positive or appetitive stimuli while they are more likely to avoid negative or aversive stimuli. For instance, in Lang's (1995) theory, the impact is driven by two primary motivational systems: an appetitive system, characterised by a behavioural approach of positive or desired stimuli, and an aversive system, characterised by behavioural avoidance of negative or undesired stimuli. In a similar vein, the hierarchical model of approach-avoidance motivation (Elliot, 2006) posits that people automatically evaluate the valence of stimuli, and these evaluations in turn activate a tendency to approach positive or beneficial stimuli and a tendency to avoid negative or harmful stimuli. The relative tendency to more quickly execute movements that result in the approach of stimuli, as compared

with the avoidance of stimuli, is typically referred to as an “approach bias.” The inverse pattern is typically referred to as an “avoidance bias.” To evaluate the degree to which an individual exhibits an approach bias, or avoidance bias, for a particular category of stimuli, the bias is computed

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across distinct categories of stimuli and the relative magnitude of the bias for each category of stimulus is assessed.

Early research into approach and avoidance biases observed that individuals were faster to execute movements that resulted in the approach of pleasant stimuli as compared with movements that resulted in the approach of unpleasant stimuli. In contrast, participants were also observed to be faster to execute movements that resulted in the avoidance of negative stimuli as compared with positive stimuli (Solarz, 1960). Likewise, Chen and Bargh (1999) demonstrated that individuals were faster at pulling a lever towards themselves when presented with positive words as compared with negative words on a computer screen, and were faster at pushing a lever when presented with negative words as compared with positive words.

In recent years, there has been a growing interest in the relationship between approach biases for food stimuli and food craving and consumption behaviour. However, recent reviews of the field show that this relationship is somewhat inconsistent (Fricke & Vogel, 2020; Kakoschke et al., 2019; Loijen et al., 2020). For instance, although Brockmeyer et al. (2015) revealed a positive association between approach bias towards food stimuli and food craving, other studies could not reliably replicate this effect (Lender et al., 2018). As a further indication of the conflicting findings, while some studies revealed that overweight and obese individuals, but not healthy-weight individuals, were faster to approach than to avoid food stimuli (Kakoschke et al., 2017b; Mehl et al., 2018), other studies could not replicate this effect and some studies even found avoidance of food in obese individuals (Paslakis et al., 2017; Schmidt et al., 2018). Given that food-related approach/avoidance biases are proposed to be causally involved in the development and maintenance of eating and weight disorders (for reviews, see Kakoschke et al., 2017a; Loijen et al., 2020), it is important that the inconsistency in findings linking such biases to food consumption is resolved.

One explanation for these inconsistent findings lies with tasks that have been used to measure approach/avoidance biases. Most studies to date have used one of two reaction time tasks: the approach-avoidance task (AAT; Chen & Bargh, 1999; Solarz, 1960) or the stimulus-response compatibility task (SRCT; De Houwer et al., 2001). In a typical AAT, participants are required to execute responses that result in either the approach, or avoidance, of stimulus images as fast as possible. Typically, responses on the AAT result in a zooming effect to simulate physical approach or avoidance, whereby approach responses are accompanied by the image increasing in size, and avoidance responses are accompanied by the image decreasing in size. Most studies have used a joystick to register responses, with participants pulling the joystick towards them to perform approach responses and pushing the joystick away from them to perform avoidance

responses. However, researchers have also used computer mouse movements (Paslakis et al., 2017) or keyboard key presses (Khan & Petróczi, 2015) to register responses. In a typical SRCT, participants are not required to execute responses that replicate approach or avoidance of stimulus images relative to themselves, but instead, participants are required to move a manikin figure either towards (representing approach) or away (representing avoidance) from image stimuli via a keyboard key press.

For both the AAT and the SRCT, instructions to approach or avoid the contrasted picture categories can involve a motivationally relevant feature of the picture (e.g., whether a picture contains chocolate or vegetables) or a motivationally irrelevant feature of the picture (e.g., whether a picture is a portrait or landscape in orientation). Irrelevant feature versions of the AAT are thought to index implicit biases more greatly because instructions lessen conscious evaluation of picture content (Lender et al., 2018; Rinck & Becker, 2007). Importantly, although both relevant and irrelevant feature versions of the AAT and the SRCT tasks have been used by researchers in the consumption domain, irrelevant feature versions typically yield smaller approach/avoidance bias estimates as compared with relevant feature tasks (Field et al., 2011; Kersbergen et al., 2015; Krieglmeier & Deutsch, 2010; Lender et al., 2018; Meule et al., 2019). For example, Lender et al. (2018) demonstrated a robust approach bias towards appetitive food images was present only under task conditions that employed relevant feature instructions and not under conditions that employed irrelevant feature instructions. In addition, researchers have reported much poorer internal reliability estimates for irrelevant feature tasks when compared with relevant feature tasks measuring approach to appetitive images (Kersbergen et al., 2015). Thus, existing research indicates that AAT and SRCT tasks that adopt instructions to approach or avoid image stimuli based on motivationally relevant features provide stronger and more reliable estimates of approach/avoidance biases across individuals.

Critically, although both the AAT and SRCT tasks are commonly employed to assess approach biases and their relation to food consumption behaviour, no research has yet directly compared the reliability and the convergence of measures yielded by AAT and SRCT task designs that have been adopted by researchers in the food consumption domain. Research that has examined these questions in other individual difference domains, such as fear and problematic alcohol consumption, has observed that measures yielded from AAT and SRCT tasks are not associated in these domains. For example, Krieglmeier and Deutsch (2010; Experiment 3) found that measures of spider avoidance bias yielded by the SRCT did not correlate with the measure yielded by the AAT across individuals who varied in fear of spiders. Similarly, Kersbergen et al. (2015) found that although alcohol approach bias scores yielded by the

Table 1. Descriptive statistics of demographic and questionnaire measures of the final participant sample, for each ABAMT task condition ($N = 121$).

Measure		
Gender	(Woman:Man) 51:70	
	Mean (SD)	Range
Age (years)	40.73 (11.60)	19–68
Body mass index (BMI)	26.49 (6.54)	17.22–58.82
Food craving questionnaire-trait scale-reduced (FCQ-T-r)	34.74 (12.69)	15–75
Dutch eating behaviour questionnaire		
Drive for thinness scale	2.62 (0.91)	1–5
Emotional eating scale	2.17 (0.94)	1–5
Externalised eating scale	2.98 (0.69)	1–4.8

SD: standard deviation.

AAT and the SRCT each explained a small proportion of variance in problematic drinking, scores yielded by the AAT and the SRCT did not correlate. Given the absence of an association between measures yielded by the AAT and SRCT tasks in these domains is consistent with the possibility that the tasks do not measure the same process, and given the frequent use of each task by researchers seeking to understand the association between approach biases and food consumption, it is imperative that the internal reliability, convergent validity, and criterion validity of measures yielded by the AAT and SRCT in the context of food consumption are investigated.

Thus, the aim of the present study was to address whether (1) the AAT and SRCT paradigms provide internally reliable estimates of approach bias to unhealthy relative to healthy food stimuli, (2) the AAT and SRCT paradigms demonstrate convergence in their assessment of approach bias to unhealthy relative to healthy food stimuli, and (3) the approach bias scores to unhealthy relative to healthy food yielded by the AAT and SRCT paradigms demonstrate associations with individual differences in food craving and eating behaviour. To do so, participants completed a mouse-based AAT, a keyboard-based AAT, and an SRCT, which have each been adopted by researchers in the food consumption domain. Given that relevant feature paradigms have been demonstrated to elicit stronger and more reliable effects, as compared with irrelevant feature paradigms (Hans Phaf et al., 2014; Lender et al., 2018), the present study employed relevant feature paradigms for each task. Each task yielded an estimate of participants' approach bias towards unhealthy food relative to healthy food. We compared the internal reliability of each of these bias scores, the degree to which the approach bias scores converged, and the degree to which the approach bias scores from each task were concurrently associated with individual differences in food craving and eating behaviour.

Method

Materials used in this study are available at <https://osf.io/pktgz/>. The in-principle accepted Stage 1 study protocol is available at <https://osf.io/kd5fn>. No data were collected before acceptance of the Stage 1 manuscript.

Participants

A sensitivity analysis determined that $r = .26$ is the smallest effect size that could be detected with 120 participants with at least 90% power ($1 - \beta = .90$) under a one-tailed test of statistical significance set at $p < .05$. This effect size is smaller than $r = .50$, commonly considered to represent a minimum acceptable level of convergent validity (Carlson & Herdman, 2012). Thus, this sample size was considered appropriate for detecting the smallest effect size of interest for the purpose of determining whether measures yielded by the assessment tasks show a clear association with questionnaire measures of individual differences in food craving and converged enough to be considered equivalent measures of the same behavioural tendencies.

Participants were recruited online via Amazon Mechanical-Turk (MTurk) in exchange for monetary compensation. Given that the stimulus set used in the study reflected foods from an omnivorous diet, participation in the study was restricted to individuals who follow an omnivorous diet. Participant recruitment continued until a cohort of at least 120 individuals had completed the study procedure and whose data were determined valid for inclusion in statistical analyses. In total, 140 individuals completed the study procedure, yielding a final participant sample of 121 valid participants who were included in statistical analysis (see *Data Screening and Participant Exclusion* section located in section "Results"). Demographic information for the final participant sample is presented in Table 1 in section "Results."

Questionnaires

Demographic and Dietary Status Questionnaire. Participants reported demographic information, including gender identity, age, dietary status (omnivorous or other), diet status (currently restricting caloric intake, or not), and current height and weight to enable computation of body mass index ($\text{BMI} = \text{kg}/\text{m}^2$).

Food Cravings Questionnaire-Trait-reduced. Dispositional tendency to experience general food cravings was assessed using the English language version of the 15-item Food Cravings Questionnaire-Trait-reduced (FCQ-T-r; Meule et al., 2014). Items were rated on a five-point Likert-type scale (1 = *strongly disagree*, 5 = *strongly agree*) with higher scores indicating higher trait food craving. The English language version of the FCQ-T-r has demonstrated excellent internal consistency reliability and good convergent validity (Hormes & Meule, 2016). The internal consistency of this measure in the current study was $\alpha_{\text{Cronbach}} = .94$, $\text{CI}_{95\%} [.93, .96]$.

Dutch Eating Behaviour Questionnaire. Trait eating behaviours were assessed using the Dutch eating behaviour questionnaire (DEBQ; van Strien et al., 1986). These eating behaviours are measured via three subscales that include *restrained eating* (10 items), *emotional eating* (13 items), and *external eating* (10 items). Respondents rate each item on a five-point Likert-type scale, ranging from 1 (*never*) to 5 (*very often*). For each subscale, an average score was computed, with higher scores indicating greater levels of engagement in restrained, emotional, or external eating behaviours. The three subscales have demonstrated excellent internal consistency, factorial validity, and predictive validity with eating behaviours (van Strien et al., 1986, 2007, 2012). The internal consistency of each subscale of this measure in the current study was as follows: *restrained eating* $\alpha_{\text{Cronbach}} = .94$, $\text{CI}_{95\%} [.92, .95]$, *emotional eating* $\alpha_{\text{Cronbach}} = .96$, $\text{CI}_{95\%} [.96, .97]$, and *external eating* $\alpha_{\text{Cronbach}} = .89$, $\text{CI}_{95\%} [.86, .92]$.

Stimulus images

Eighty images displaying food items were selected from the Extended Food-pics Database (Blechert et al., 2019) for use in the approach bias assessment tasks. Half of the images (40) depicted healthy foods, while the other half (40) depicted unhealthy foods. To objectively differentiate healthy and unhealthy foods, two key nutritional components (i.e., sugar and fat content $\text{gram}/100\text{g}$) were estimated for each food image using the FoodData Central Database of the United States Department of Agriculture (<https://fdc.nal.usda.gov/>). In line with Australian Dietary Guidelines (National Health and Medical Research Council, 2013), a food item was classified as healthy if its

sugar and fat content did not exceed 10/100 g, unless most of the sugar was coming from fruit, and unhealthy if it did not meet this criterion. As expected, this criterion resulted in unhealthy food images depicting foods that are, on average, significantly higher in calorie content than foods depicted in healthy food images, $t(40.48) = 3.73$, $p < .001$. Healthy and unhealthy foods did not differ in brightness, complexity, and recognisability (all $ps > .06$). An additional set of 12 images was selected for use in practice trials of the approach bias assessment tasks.

Approach bias assessment tasks

Participants completed three approach bias assessment tasks that reflected task designs used in the investigation of the relationship between approach biases and craving and eating behaviour. These tasks were the SRCT, the AAT—Arm Movement Response, and the AAT—Key Press Response. Features that are common across these tasks are described first, followed by a description of those features unique to each task.

Features common across approach bias assessment tasks. Each approach bias assessment task consisted of two blocks presented in a fixed order. The first block, labelled the *approach unhealthy foods block*, presented trials in which participants were required to execute task-specific responses that resulted in approach towards unhealthy foods and avoidance of healthy foods. The second block, labelled the *approach healthy foods block*, presented trials in which participants were required to execute task-specific responses that resulted in an approach towards healthy foods and avoidance of unhealthy foods. Each assessment block contained 80 trials and was immediately preceded by a block of instructions and 12 practice trials that provided feedback on the correctness of each response. Images used in each block were derived from the Assessment Stimulus Set described earlier, with each image randomly presented once per block. All trials were presented on a black background.

For each participant at the conclusion of each task, an index of behavioural approach was computed by subtracting the median response latency of correct approach trials from the median response latency of correct avoidance trials, separately for trials depicting unhealthy foods and for trials depicting healthy foods. Thus, each index reflected the relative speeding for participants to execute responses that approach each food type, relative to responses that avoid each food type. Next, an *Unhealthy Food Approach Bias Index* score was computed by subtracting the latter value (the degree of speeded approach for healthy foods) from the former value (the degree of speeded approach unhealthy foods). Thus, larger index scores on the *Unhealthy Food Approach Bias Index* represented a larger bias favouring a more speeded approach, relative to

avoidance, of unhealthy foods as compared with healthy foods. This computation can be expressed as:

$$\text{Unhealthy Food Approach Bias Index} = [\text{Median latency for avoiding unhealthy foods} - \text{Median latency for approaching unhealthy foods}] - [\text{Median latency for avoiding healthy foods} - \text{Median latency for approaching healthy foods}]$$

Stimulus-Response Compatibility Task. Each trial of the SRCT started with a 1,000-ms black screen. Next, a single picture (80 mm × 80 mm) was presented in the centre of the screen, and a white stick figure manikin (55 mm × 55 mm) was presented either above or below the picture. In approach unhealthy foods blocks, participants were required to move the manikin towards images containing unhealthy foods and away from images containing healthy foods, using either the Y-key (move manikin up) or the B-key (move manikin down) of a standard QWERTY keyboard. In approach healthy foods blocks, participants were required to move the manikin towards images containing healthy foods and away from images containing unhealthy foods, using the same response keys. Across all blocks, the four different trial types following from combinations of the manikin's starting position (above versus below the image) and the image content (healthy versus unhealthy food) were presented in random order but with equal frequency. Upon a correct response, the manikin moved in the appropriate direction. An incorrect response was immediately followed by a 5-second delay with the text "incorrect response" displayed on the screen. The time taken for participants to execute the key press response following the presentation of the image, and accuracy of the response, was recorded by the task.

AAT-arm movement response. Each trial of the AAT with arm movement response started with a 1,000-ms black screen. Next, a red "X" was presented in the middle of the screen. Participants were required to click on the X with their mouse pointer to initiate each trial. Immediately following, the "X" was replaced by a square picture sized to be 50% of the screen height. In approach unhealthy foods blocks, participants were instructed to execute an approach movement by contracting their arm to pull the mouse towards themselves in response to images containing unhealthy foods, and to execute an avoidance movement by extending their arm to push the mouse away from themselves in response to images containing healthy foods. In approach healthy foods blocks, participants were instructed to execute the approach movement in response to images containing healthy foods and execute the avoidance movement in response to images containing unhealthy foods. To simulate the change in the physical distance between the picture and participant, the size of the image increased or decreased in response to the movement of the mouse. Specifically, when participants pulled the mouse towards

themselves the picture progressively increased in size and when participants pushed the mouse away from themselves the picture progressively reduced in size. The trial concluded when the image had either increased to the total height of the screen or had reduced to 10% of the total height of the screen. Trials on which participants made incorrect responses were immediately followed by a 5-s delay with the text "incorrect final mouse movement" displayed on screen. At the conclusion of each trial, the time taken for participants to complete the mouse movement response following the presentation of the image, and accuracy of the response, was recorded by the task.

AAT-key press response. Each trial of the AAT with key press response was identical to the design of the arm movement response task, except participants were required to execute responses using a keyboard key press rather than the movement of the mouse. Specifically, in approach unhealthy foods blocks, participants were required to execute an approach response by pressing the down-arrow key on the keyboard when images containing unhealthy foods appeared on the screen, and to execute an avoidance response by pressing the up-arrow key on the keyboard when images containing healthy foods appeared on the screen. In approach healthy foods blocks, participants were required to execute the approach response when images containing healthy foods appeared on the screen and execute the avoidance response when images containing unhealthy foods appeared on the screen. To simulate change in the physical distance between the picture and participants, the size of the image increased or decreased in response to the key press. Specifically, when participants pressed the down-arrow key, the picture progressively increased in size, and when participants press the up-arrow key, the picture progressively reduced in size. An incorrect response was immediately followed by a 5-s delay with the text "incorrect response" displayed on the screen. At the conclusion of each trial, the time taken for participants to execute the key press response following the presentation of the image, and accuracy of the response, was recorded by the task.

Procedure

All procedures necessary to conduct this research were approved by the Human Research Ethics Committee at the University of Western Australia. The procedure was delivered via the internet to participants' personal computers using the Inquisit Web experiment platform. Before recruitment, individuals were informed that the study seeks to examine patterns of cognitive processing across different assessment conditions and were informed of the requirements involved in participation. Upon accepting the invitation to participate in the study participants were directed to a URL that hosts the experiment assessments.

Table 2. Descriptive statistics of measures of Unhealthy Food Approach Bias Index scores and total response accuracy recorded from the Approach Avoidance Task and Stimulus-Response Compatibility Task assessments; $N = 121$.

Assessment Task	Index Measure	Index Score (ms)		Response Accuracy (prop. correct)	
		<i>M (SD)</i>	<i>Range</i>	<i>M (SD)</i>	<i>Range</i>
AAT	Unhealthy Food Approach Bias Index	-118.74 (229.20)	-893.00 to 820.50	.87 (.068)	.68–.99
Arm Movement Response Task	<i>Speeded approach—Unhealthy foods</i>	-48.72 (176.56)	-547.00 to 1,063.00		
	<i>Speeded approach—Healthy foods</i>	70.02 (115.91)	-224.50 to 595.50		
AAT	Unhealthy Food Approach Bias Index	-53.64 (217.22)	-928.50 to 794.00	.95 (.034)	.86–1
Keyboard Response Task	<i>Speeded approach—Unhealthy foods</i>	-18.10 (138.05)	-466.00 to 520.00		
	<i>Speeded approach—Healthy foods</i>	35.24 (118.34)	-310.50 to 462.50		
SRCT	Unhealthy Food Approach Bias Index	-258.18 (414.72)	-2,082.50 to 1,400.00	.94 (.043)	.76–1
	<i>Speeded approach—Unhealthy foods</i>	-86.02 (255.24)	-1,265.00 to 845.50		
	<i>Speeded approach—Healthy foods</i>	172.16 (202.59)	-554.50 to 817.50		

AAT: approach-avoidance task; SD: standard deviation; SRCT: stimulus-response compatibility task.

Note: Bold values represent the final index measure (double difference index) for each task, while non-bold values reflect the constituent simple difference scores. Positive values for speeded approach measures reflect relatively speeded approach responses as compared to avoidance responses, while negative values reflect relatively speeded avoidance responses as compared to approach responses. Data for all task conditions are available in the public data repository.

Participants were first required to provide informed consent and demographic information before then completing the questionnaires. Participants next completed the approach bias assessment tasks. Tasks were delivered in a counterbalanced order such that an equal number of participants completed the tasks in each possible order. The tasks were programmed to maintain equivalent spatial parameters among visual stimuli across different screen sizes and resolutions. The length of the procedure was anticipated to be less than 60 min. Upon completion of the tasks, participants were debriefed.

Results

Data and analysis scripts are available at <https://osf.io/pktgz/>.

Data screening and participant exclusion

Participants were excluded from all analyses if they met low accuracy exclusion criteria as follows. Participants were first excluded if they demonstrated a response accuracy rate of less than 50% on any assessment task (three participants), and subsequently, if they demonstrated a response accuracy rate that was more than 2.58 standard deviations below the mean rate across all remaining participants on any assessment task (nine participants). Next,

to eliminate the influence of extreme data points upon analyses, any participant who demonstrated an Unhealthy Food Approach Bias Index further than 3 *SDs* from the participants' sample mean on any assessment task was excluded from further analyses (seven participants). Thus, the final data set included the data of 121 participants.

Statistical analyses

Descriptive statistics. Descriptive statistics concerning participants' demographic and questionnaire measures are presented in Table 1. Statistics describing participants' response accuracy and Unhealthy Food Approach Bias Index scores for each task are presented in Table 2.

Internal reliability of bias index scores. A measure of internal consistency was computed for each assessment task using a random permutation split-half correlation approach (Parsons, 2021). This approach splits each participant's trial data into two randomly selected halves before computing the Unhealthy Food Approach Bias Index for each half. Pearson correlation coefficients are then computed between these indices across all participants, including the Spearman–Brown (SB) formula adjustment. This process was then repeated across 5,000 randomly sampled split-half trial combinations. A measure of internal consistency was then determined by computing the mean and 95%

Table 3. Correlations between task and questionnaire measures. (N = 121).

		1	2	3	4	5	6	7
1.	AAT-arm movement response task	—						
2.	AAT-keyboard response task	.16* $p = .040$	—					
3.	SRCT	.29*** $p < .001$	.10 $p = .15$	—				
4.	FCQ-T-r	.17* $p = .030$	.04 $p = .32$	-.06 $p = .76$	—			
5.	DEBQ—drive for thinness scale	-.09 $p = .84$	.06 $p = .26$	-.03 $p = .64$	.10 $p = .15$	—		
6.	DEBQ—emotional eating scale	.10 $p = .14$	.00 $p = .49$	-.10 $p = .86$	.68*** $p < .001$	.15* $p = .049$	—	
7.	DEBQ—externalised eating scale	.18* $p = .022$	.00 $p = .49$	-.05 $p = .69$	.64*** $p < .001$	.06* $p = .042$	.61*** $p < .001$	—

Note: AAT: approach-avoidance task; DEBQ: Dutch eating behaviour questionnaire; FCQ-T-r: food craving questionnaire-trait scale-reduced; SRCT: stimulus-response compatibility task. Results reflect Spearman rank-order correlations conducted using a one-tailed test of statistical significance for the presence of a positive correlation. Bold values represent statistically significant effects.

* $p < .05$; ** $p < .01$; *** $p < .001$.

confidence interval (CI) of the distribution of Pearson correlation coefficients observed across the sampled split-half trial combinations.

These analyses revealed that the internal consistency of the Unhealthy Food Approach Bias Indices yielded from each of the behavioural tasks was as follows: AAT-Arm Movement Response Task $r_{SB} = .79$, $CI_{95\%} [.64, .87]$; AAT—Key Press Response Task $r_{SB} = .77$, $CI_{95\%} [.67, .84]$; SRCT $r_{SB} = .87$, $CI_{95\%} [.80, .92]$.

Convergence of bias index scores. Correlational analysis was employed to examine convergence between the Unhealthy Food Approach Bias Index scores yielded by each assessment task to determine whether these assessment tasks plausibly measured the same approach action tendencies. In all cases, variables were not normally distributed and so associations were examined using Spearman's Rho correlations. Given the presence of clear directional hypotheses predicting positive associations between these measures, statistical significance for these analyses was determined using one-tailed tests with alpha-criterion of $p < .05$. Correlations between these task measures are reported in Table 3. These analyses revealed small statistically significant correlations between the AAT-Arm Movement Response Task and both the AAT-Key Press Response Task and the SRCT. No significant correlation was found between the AAT-Key Press Response Task and the SRCT.

Associations between bias index scores and questionnaire measures. Correlational analyses were employed to determine whether the Unhealthy Food Approach Bias Index scores yielded by each assessment task were associated with individual differences in self-reported food craving (FCQ-T-r) and eating behaviours (DEBQ subscales). In all cases, at least one variable was not

normally distributed and so associations were examined using Spearman's rho correlations. Given the presence of clear directional hypotheses predicting positive associations between these measures statistical significance for these analyses were determined using one-tailed tests with alpha-criterion of $p < .05$. Correlations between these task measures are reported in Table 3. These analyses revealed small statistically significant correlations between the Unhealthy Food Approach Bias Index score yielded by the AAT-Arm Movement Response Task and the External Eating Subscale of the DEBQ. No other correlations between Unhealthy Food Approach Bias Index scores and questionnaire measures were statistically significant.

Discussion

The present study aimed to address whether (1) the AAT and SRCT paradigms provide internally reliable estimates of approach bias to unhealthy relative to healthy food stimuli, (2) the AAT and SRCT paradigms demonstrate convergence in their assessment of approach bias to unhealthy relative to healthy food stimuli, and (3) the approach bias scores to unhealthy relative to healthy food yielded by the AAT and SRCT paradigms demonstrate associations with individual differences in food craving and eating behaviour. Participants completed relevant feature versions of the mouse-based AAT, keyboard-based AAT, and SRCT tasks. Each task yielded an estimate of participants' approach bias towards unhealthy food relative to healthy food. Analyses assessed the internal reliability of these bias scores, the degree the approach bias scores from each task converged, and the degree the approach bias scores from each task were associated with individual differences in self-reported food craving and eating behaviour.

Although measures of internal reliability of each assessment task showed the SRCT task to yield nominally higher intercorrelation as compared with the AAT variants, the examination of CIs showed significant overlap in the reliability estimates across all three tasks. Thus, the present study demonstrated the AAT-Arm Movement Response Task, AAT-Key Press Response Task, and SRCT task to hold equivalent levels of internal reliability. Although measures of internal reliability are highly subject to interpretation, guidelines for the interpretation of such measures have commonly suggested intracorrelation coefficients of $r \geq .70$ represent acceptable levels of internal consistency (Ponterotto & Ruckdeschel, 2007). Under such guidelines, the internal consistency of measures yielded by each assessment task was acceptable.

Analyses also revealed evidence of some convergence between Unhealthy Food Approach Bias Index scores yielded by the AAT-Arm Movement Response Task and both the AAT-Key Press Response Task and the SRCT task. However, the strength of association between the index scores yielded by the AAT-Arm Movement Response Task and the other two tasks was modest, indicating the measures reflect a considerable proportion of variance that is not shared between the tasks. Importantly, no other statistically significant associations were observed between the index scores yielded by the three assessment tasks. This indicates that variation in bias index measures yielded by the AAT-Key Press Response Task and the SRCT task did not align across individuals in the present study. Thus, the present study demonstrated that convergence between the three assessment tasks was inconsistent, such that index scores used to measure biased approach towards unhealthy foods in the AAT-Key Press Response Task and the SRCT task each converged modestly with index scores from the AAT-Arm Movement Response Task, but not with each other.

Finally, results revealed evidence of positive associations between Unhealthy Food Approach Bias Index scores yielded by the AAT-Arm Movement Response Task and questionnaire measures of external eating behaviours and trait food craving. However, no association was observed between AAT-Key Press Response Task and the SRCT task measures and any questionnaire scale. Thus, in the present sample, individuals who reported relatively higher levels of external eating behaviours and trait food craving were characterised by more speeded execution of arm movements that favour approach towards, rather than avoidance of, unhealthy foods as compared to the same arm movements executed in relation to healthy foods. However, when tasks did not involve arm movements, no association was observed between any questionnaire measure of eating behaviour or food craving.

The internal reliability estimates of the task AAT-Arm Movement Response Task observed in the present study ($r_{SB} = .79$, $CI_{95\%} [.64, .87]$) is consistent with estimates

obtained in recent studies that examined the internal reliability of both joystick ($r_{SB} = .76$) and mouse ($r_{SB} = .75$)-based response modes in feature-relevant versions of the AAT paradigm in the food consumption domain (Wittekind et al., 2021). These findings suggest that the reliability of the AAT-Arm Movement Response Task may be stable across studies, response modes, and stimulus image sets. Although we were unable to identify studies that have examined the internal reliability of the AAT-Key Press Response Task or the SRCT task in the food consumption domain, studies using these tasks in other appetitive domains reported slightly smaller split-half correlations. For instance, Field et al. (2011) found that an SRCT in the context of alcohol consumption yielded a split-half reliability of $r_{SB} = .63$, and Kahveci et al. (2020) found that an AAT-Key Press Response Task in the context of pornography consumption yielded a split-half reliability of $r_{SB} = .52$. In sum, our present findings suggest that all three tasks included in the study exhibit acceptable levels of internal reliability to assess approach bias in the food consumption domain. We would like to encourage researchers to report the internal reliability of approach-avoidance assessment tasks to allow for a comparison of the stability of these measures across studies.

The present findings demonstrate that approach bias index scores from common approach-avoidance assessment tasks show inconsistent evidence of convergence with one another. However, it is noteworthy that, within the same participant sample, measures from each task demonstrated acceptable levels of internal reliability alongside clear evidence of variance within the sample, indicating that in each case these tasks reliably measured some source of individual difference. Given this, the present lack of systematic convergence between tasks indicates that they do not measure the same sources of individual difference with respect to biased action tendencies in the food consumption domain. Of most interest is the observation that though measures from the AAT-Key Press Response Task and the SRCT task did not converge with one another, they each converged with measures from the AAT-Arm Movement Response Task. This indicates that the measure yielded by the arm movement response modality may incorporate distinct cognitive or behavioural processes reflected in the measures of the other two tasks.

The current study observed that only Unhealthy Food Approach Bias Index scores yielded by the AAT-Arm Movement Response Task, and not index scores yielded by either the AAT-Key Press Response Task or the SRCT task, were associated with self-report measures of eating behaviour and food craving. These results suggest that the AAT with arm movement response modality was the only measure sensitive to detecting biased approach-avoidance behaviour linked to eating behaviour and craving in the food consumption domain. The finding supporting the suitability of arm movement response modalities in the

AAT paradigm is consistent with researchers' arguments for the importance of sensorimotor response modes (e.g., arm movements) to revealing approach-avoidance effects (Dondzilo et al., 2021; Meule et al., 2020; Moore et al., 2022), as such responding more closely simulates behaviours during food consumption. In support of this notion, and consistent with the current findings, evidence for an association between measures of approach bias and food craving has been shown in studies using AAT paradigms with arm movement response modes (Brockmeyer et al., 2015; Lender et al., 2018; Meule et al., 2019) but not in studies that have used SRCT tasks (Knight et al., 2020; but see Kahveci et al., 2020, 2021, e.g., and potential explanations for, touchscreen-based AAT designs that have not evidenced associations between approach bias indices and food craving). Nonetheless, the present study has the unique power of examining participants responding to arm-movement and non-arm-movement response mode tasks in response to the same stimuli and within the same sample of individuals, which demonstrated the AAT with arm movement response mode to be uniquely capable of detecting associations between food-related approach biases and individual differences in food craving and external eating behaviours, compared with the SRCT task or an AAT task with a keyboard response mode.

It is important to consider some aspects of the present study's design when interpreting the present findings. The present study only provides an evaluation of the assessment tasks under conditions that compare responding to images of healthy and unhealthy foods and examined associations of task measures with measures of eating behaviour and food craving. The reliability and convergence of identical paradigms employed in other domains, such as drug and alcohol consumption, fear, emotion, or body image, may yield different patterns of findings. Indeed, Krieglmeier and Deutsch (2010) found that measures of biased avoidance of spider stimuli yielded by the SRCT did not correlate with a measure yielded by an arm-movement AAT across individuals who varied in fear of spiders. Similarly, Kersbergen et al. (2015) observed that scores yielded by arm-movement AAT and an SRCT did not correlate with one another across individuals who varied in problematic drinking. Furthermore, the present study employed versions of the AAT and SRCT tasks that require participants to respond to features of the stimulus images that are relevant to their content (i.e., the "relevant feature" paradigm). It is possible that the measurement reliability, convergence, and sensitivity of each task may be altered when participants are required to respond to features of the stimulus images that irrelevant to their content (i.e., the "irrelevant feature" paradigm). However, previous studies have reported discouragingly low-reliability estimates for such irrelevant feature approach/avoidance paradigms (e.g., Field et al., 2011; Kersbergen et al., 2015; Meule et al., 2019). Finally, the study design utilised an unselected

sample of participants who varied in food consumption behaviours and craving. Although such approaches have the benefit of revealing associations between measures across their distribution and more accurately represent the size of associations, extreme group designs have the benefit of maximising the opportunity to detect differences in measures at extreme ends of individual difference distributions, particularly when effect sizes are small (Preacher, 2015). Furthermore, although the participant recruitment strategy did result in the recruitment of individuals who varied widely in eating behaviour and craving, the sample contained relatively few individuals who reported the level of craving and eating behaviour that would be deemed clinically relevant (Meule, 2018; Wardle, 1987). Given this, it is possible that the sensitivity of each task to detect differences in task responding linked to consumption and craving may be enhanced when comparing groups of individuals who differ greatly in their patterns of eating behaviour and craving.

For the moment, however, the present study has demonstrated several characteristics of variants of the AAT and SRCT when used to measure biases favouring approach towards unhealthy foods among people who vary in eating behaviour and trait craving. The study demonstrated that measures yielded by these tasks exhibit comparable and acceptable levels of internal consistency, that the tasks yielded modest or absent levels of measurement convergence, and that only the AAT task using an arm movement response modality demonstrated associations between its measure of biased approach to unhealthy foods and self-report measures of eating behaviour and trait food craving. Together these findings suggest that measures from each task do not converge strongly enough to be considered equivalent measures of approach-avoidance biases, and that the AAT task using an arm movement response modality may be uniquely suited to detecting biases in approach or avoidance of foods believed to characterise individuals with elevated maladaptive eating behaviour and craving. It is hoped that these findings will usefully inform researchers who seek to investigate approach-avoidance processes in the food consumption domain.

Author contributions

All authors contributed equally to the work described in this manuscript and its publication.

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Data accessibility statement



The data and materials from the present experiment are publicly available at the Open Science Framework website: <https://osf.io/pktgz/>

Publication

This is a Stage 2 Registered Report manuscript. Stage 1 of this manuscript has passed peer review and has been issued an in-principle acceptance (IPA) by the Quarterly Journal of Experimental Psychology, indicating that the article will be published pending successful completion of the study according to the exact methods and analytic procedures outlined, as well as a defensible and evidence-bound interpretation of the results.

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